W-B-101J
November 19, 1982
SUPERSEDING
W-B-101H
June 29, 1973

FEDERAL SPECIFICATION

BATTERY, DRY

This specification is approved by the Assistant Administrator, Office of Personal Property, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers primary (nonrechargeable) dry unitcells and batteries having a nonspillable electrolyte.

1.2 Classification

1.2.1 Classes. Dry unitcells and batteries furnished under this specification shall be of the following classes and battery designations, as specified (see table II and 6.1).

Classes:

1 No. 6 - General Purpose, Cells
2 No. 6 - Industrial, Cells
3 No. 6 - Regular Telephone, Cells
4 No. 6 - Special Telephone, Cells
5 No. 6 - Alarm, Cells
6 - Assembled batteries of No. 6 cells
7 - Group batteries of small cells intended for No. 6 application
8 - Lantern batteries
12 - Heavy Duty flashlight cells
13 - Photographic unitcell batteries
14 - Transistor hearing-aid unitcells and batteries
16 - High-voltage batteries (formerly B or C batteries), 15V and over
17 - Transistor radio and electronic equipment batteries
18 - Watch cells
19 - High-rate primary cells
20 - Miniature hearing-aid cells and batteries
22 - Toy batteries
23 - Motor batteries
24 - Calculator batteries
25 - Smoke detector batteries

Classes 10, 11, and 21 are superseded by class 12
Classes 9 and 15 are superseded by class 8

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard

Federal Standard:

FED. STD. No. 123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 6135

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, MA; New York, NY; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc., (ANSI):

C18.1 - Specifications for Dry Cells and Batteries

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW Washington, DC 20036.)

Uniform Classification Committee, Agents:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Certification. Each manufacturer or distributor shall furnish certified test reports acceptable to the contracting officer that their batteries meet the respective tests in ANSI C18.1 for each battery designation. The solicitation shall designate on which battery a certified test report is required. The certified test report shall be dated within one year previous to the solicitation date.

3.2 General. Except as otherwise provided herein, the cells and batteries covered by this specification shall conform to the requirements of ANSI Standard C18.1 for size (see 4.3.3.1 and ANSI C18.1 table I), designation (see table I), and open circuit voltage (see 4.3.3.2).

3.3 Jackets. Jackets shall be in accordance with ANSI C18.1. When specified (see 6.1), the jacket shall be coated with parafin or a high temperature plastic sealing compound, applied to the nonconducting components for waterproofing and corrosion resisting purposes. Terminals or connectors shall not be obstructed, isolated, or insulated by such coatings (see 4.3.3.1).

3.4 Material. All materials used in the construction of the finished product shall conform to the applicable requirements of this specification. Where materials are not defined, the unit products shall be of the quality normally used for the purpose in commercial practice.

3.5 Terminals. Unless otherwise specified (see 6.1), the terminal location, shape, type, and dimensions shall conform to the requirements of ANSI Standard C18.1 for the cell or battery designation indicated.

3.6 Performance requirements. The dry cells and batteries shall meet the applicable performance requirements of ANSI C18.1 for the Unitcell or Battery (see 4.3.2 and table I).

3.7 Marking. Marking shall be in accordance with ANSI C18.1.

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3.8 Designation. Battery designation per class shall be in accordance with table II.

3.9 Date designation. A letter "D" shall be used to designate each battery that will have a specific numerical date (month, day, year) of manufacture. These batteries will be designated in the solicitation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Inspections and tests. The inspection and testing of the dry cells and batteries covered by this specification shall be as specified in 4.3.

4.3 Lot acceptance inspections, and tests.

4.3.1 Inspection lot. All dry cells and batteries of one class and size presented at one time shall be considered a lot for purposes of lot acceptance.

4.3.2 Samples. Sample cells and batteries shall be selected in accordance with MIL-STD-105, Inspection Level S-2. The acceptable quality level (AQL) shall be 2.5.

4.3.3 Inspection. The samples selected in accordance with 4.3.2 shall be subject to the inspections and tests in 4.3.3.1 and 4.3.3.2 to determine compliance with the requirements of this specification.

TABLE I. Certification

Cell or battery class No.	Applicable tables in ANSI Standard C18.1		
	Cell or Battery designation	Performance requirements	Sizes
Class			
1. General purpose cells No. 6	3	3	1
2. Industrial cells No. 6	3	3	1
3. Telephone cells (regular) No. 6	3	3	1
4. Telephone cells (special) No. 6	3	3	1
5. Alarm cells No. 6	3	3	1
*6. Assembled batteries of No. 6 cells	2	3	2
*7 Group batteries of small cells for No. 6 applications	2	3	2
8. Lantern batteries	2	4	2
12. Heavy duty flashlight unitcell batteries	6	6	1
13. Photographic unitcell batteries	7	7	1
14. Transistor hearing-aid unitcells, and batteries	8	8	2
16. High-voltage multicell batteries (formerly B or C batteries), 15 V and over	10	10	2
17. Transistor radio and electronic equipment batteries	11	11	1 and 2
18. Watch unitcell batteries	12	12	1
19. High-rate unitcell batteries	13	13	1
20. Miniature hearing-aid unitcell batteries	9	9	1
22. High-rate and toy unitcell batteries	14	14	1
23. Motor unitcell batteries	15	15	1
24. Calculator batteries	16	16	1 and 2
25. Smoke-detector batteries	17	17	1 and 2

*To be certified in accordance with General Purpose Cells No. 6 Tests

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TABLE II. Battery designations per class per ANSI Standard C18.1*

Class	Battery Designations
1	905***
2	911***
3	906***
4	----
5	906***
6	907, 912, and 922***
7	903, 918, 902 and 918C***
8	908, 915C, 908C, 908D, 924, and 925***
12	AA, C, D, L40, L70, L80 and L90**
13	L15, AAA, L30, D, C, AA, L90, L70 and L40**
14	M5, M6, S6, M10, S11, M11, M12, M15, S15, M20, M25, M30, N and M35**
16	220, 221, 215, 210, 225, 201, 200 and 204***
17	15, 15M, 15A, 14, 14A, 14D, 13, 13A, 13D, 1604, 1604M, 1604A and 1604D***
18	WM5, WS6, WM8, WM10, WS10, WM15 and WS16**
19	15A, 14A, and 13A***
20	M4, S5, M5, M6, S6, M3, S11, M11, M15, and S15**
22	15, 15A, 14, 14A, 13 and 13A***
23	15, 15A, 14, 14A, 13, and 13A***
24	15, 15A, 24, 24A, 1604, 1604D, and 1604A***
25	1604A and 15A***

* All batteries included in ANSI Std. C18.1 are not included in table II, but are not precluded from use in W-B-101.
** ANSI Battery designation from ANSI Std. C18.1.
*** NEDA Battery designation from ANSI Std. C18.1.

4.3.3.1 Visual inspection. Each sample shall be visually examined for compliance with 3.3, 3.5, 3.7, and dimensionally examined for compliance with the size requirements of 3.2.

4.3.3.2 Open circuit voltage. After completion of the inspections required in 4.3.3.1, each sample shall be subjected to the voltage test set forth in ANSI Standard C18.1 to determine compliance with the open circuit voltage requirements of 3.2.

4.4 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of Section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or commercial as specified (see 6.1).

5.1.1 Level A. Drycell and batteries, in quantities as specified in 6.1, shall be packaged in close fitting paperboard or fiberboard boxes. The boxes shall be secured in a manner that will prevent loss or damage during shipment, handling, storage, and redistribution. Alternatively, the batteries shall be blister packaged in accordance with the manufacturer's standard commercial practice, providing that this will assure the degree of protection described herein.

5.1.2 Level B.

5.1.2.1 Unit packaging. The number of drycells or batteries shown in table III column 2, shall be packaged, as specified in 5.1.1.

5.1.2.2 Intermediate packaging. Where applicable, unit packages of drycells or batteries in quantities as shown in table III column 3, shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with the Appendix to PPP-B-636.

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5.1.3 Commercial. The packaging and packing of drycells and batteries shall be in accordance with normal commercial practice and shall assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.1).

5.2.1 Level A. Drycells or batteries packaged as specified in 5.1 shall be packed in quantities specified in 6.1 in a close fitting box conforming to PPP-B-636, class weather resistant, style optional. The boxes shall be closed, waterproof sealed, and reinforced in accordance with the Appendix to PPP-B-636.

5.2.2 Level B. Drycells or batteries packaged as specified in 5.1, shall be packed in quantities shown in table III in a close fitting box conforming to PPP-B-636, class domestic, style optional. The box shall be closed in accordance with the Appendix to PPP-B-636.

5.2.3 Commercial. Drycells or batteries packaged as specified in 5.1, shall be packed in a close fitting corrugated fiberboard box conforming to Section 6(a), rule 41 of the Uniform Freight Classification, and rule 222 of the National Motor Freight Classification. Closure of the boxes shall be in accordance with Section 7 of rule 41 of the Uniform Freight Classification.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

TABLE III. Quantities per unit, intermediate, and shipping container

Type of battery	No. of items per unit package	No of unit package per intermediate package	No. of intermediate or unit package per shipping container	No. of batteries to the shipping container
General purpose No. 6 battery	*	*	12	12
Size NEDA 14(C) flashlight battery	24	*	8	192
Size NEDA 15(AA), flashlight battery	24	8	8	1536
Lighting battery NEDA 908 6-volt	*	*	25	25
Telephone type battery No. 6	*	*	*	12
Size D flashlight battery High voltage batteries	24	*	8	192
NEDA 215 (15F20)	12	*	4	48
NEDA 204(30AA)	*	*	*	6
NEDA 202 (30F90)	*	*	*	6
NEDA 200 (45F40)	*	*	*	12

*Unit or intermediate packaging is nor required

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Class and designation of unitcell.
(c) Cell or battery size and voltage, if other than specified (see 3.2).
(d) Terminal arrangement, if other than specified (see 3.5).
(e) Marking of cell battery (see 3.7).
(f) Inspection responsibility, if other than specified (see 4.1).
(g) Jacket coating, if required (see 3.3).
(h) Whether civil or military marking is required (see 5.3).
(i) Number of drycells or batteries per container (see 5.1.1 and 5.2.1).

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MILITARY INTERESTS:

Custodians

Air Force - 99

Review Activities

Army - ME, MD, MI, EL
Air Force - 80
Navy - SH

User Activities

Army - AT
Navy - MC, OS, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - OPP
DC GOVT - DCG
DOT - ACO, FHW
HHS - FDA
USDA - AFS

PREPARING ACTIVITY:

GSA - OPP

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